

EEE207 ANALOG CIRCUITS PHASE ONE PROJECT REPORT BY GROUP 03



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Course Name: Analog Circuits

Course Code: EEE207

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Aim:

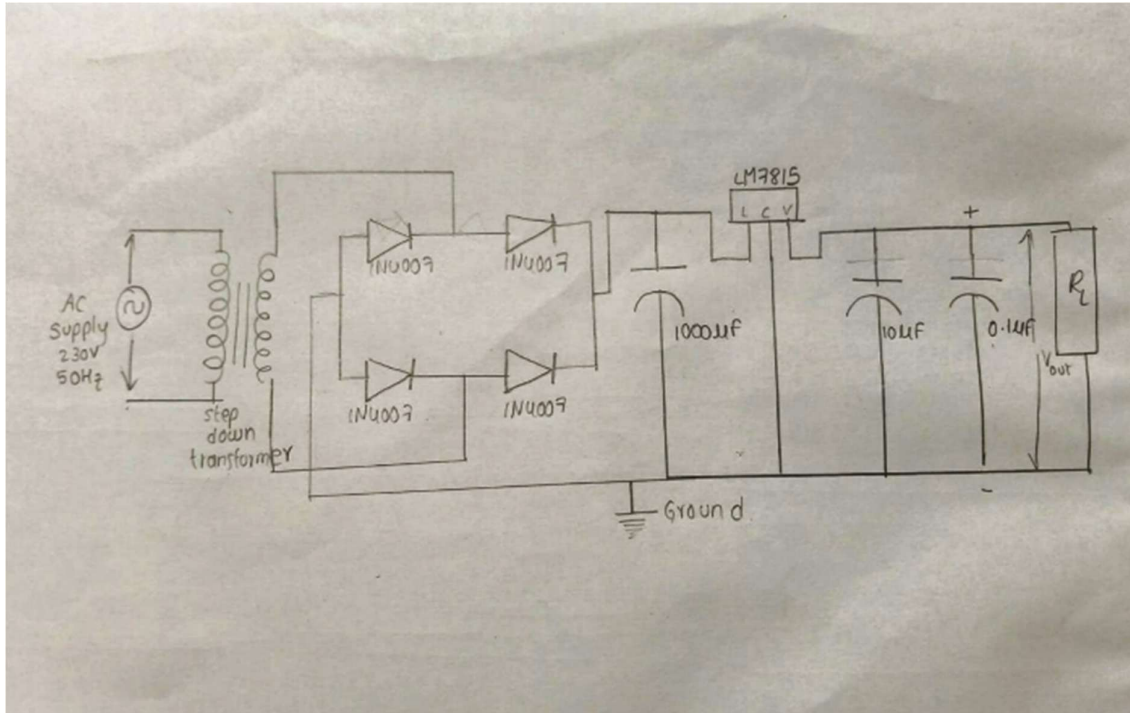
To design and simulate a regulated power supply that converts 240 V AC into a constant +15 V DC using a transformer, bridge rectifier, filter capacitor, and LM7815 voltage regulator.

Components Used:

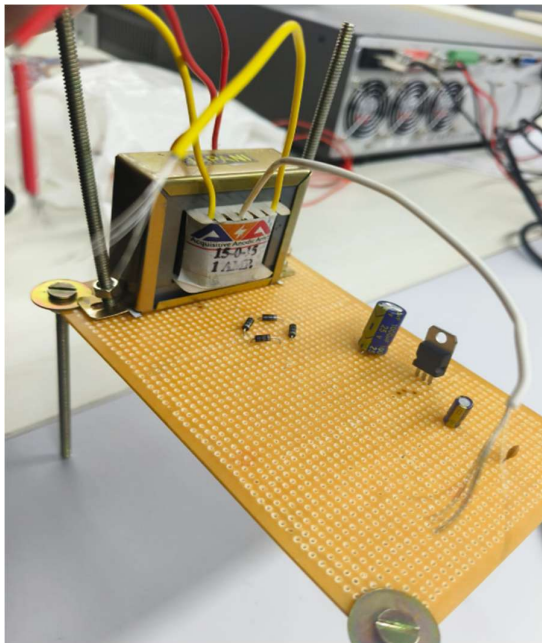
S.No	Name of the Component	Rating Specification /	Quantity
01.	Step down Transformer	230V to 15v	01
02.	Diodes	IN4007	04
03.	Capacitor	1000 μ F	01
04.	Capacitor	10 μ F	01
05.	Capacitor	0.1 μ F	01
06.	Voltage Regulator	LM7815	01
07.	PCB Board	-	01
08.	Patch cords	-	As required

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Circuit diagram:



Real time circuit connection:



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Theory and Procedure:

Transformer

We use the step down transformer to step down the 230 voltage ac source to 15v Ac which then passed to the bridge rectifier.

Bridge Rectifier

In the Bridge rectifier the four IN4007 diodes are connected as shown in the circuit diagram, this bridge rectifier convert Ac source to pulsating Dc

Filter Capacitor:

- **1000 μ F capacitor** The 1000 micro farad capacitor is used to reduce the ripples and provide smooth dc voltage
- **The 0.1 μ F capacitor** The second capacitor which is 0.1 micro farad capacitor which keeps the output voltage stable
- **The ceramic capacitor** The ceramic capacitor is used to cancel the high frequency noise and also reduce voltage spikes in the given system.

Voltage Regulator (LM7815)

The lm7815 is a positive voltage regulator that regulates the voltage and gives =15v to output

Input Cable

Through the cable we take the input ac voltage to the transformer which later passed to the bridge rectifier.

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Design Calculations:

Minimum input voltage required for LM7815:

$$V_{in(min)} = V_{out} + 2V$$

$$V_{in(min)} = 15 + 2 = 17V$$

Select transformer secondary voltage = 15V AC

Peak voltage:

$$V_{peak} = V_{rms} \times \sqrt{2}$$

$$V_{peak} = 15 \times 1.414 = 21.21V$$

Voltage drop across bridge rectifier (2 diodes):

$$V_{diode} = 0.7 \times 2 = 1.4V$$

DC voltage after rectification:

$$V_{DC} = 21.21 - 1.4 = 19.8V$$

Since $19.8V > 17V$, the transformer selection is suitable.

$$P = (V_{in} - V_{out}) \times I$$

$$P = (19.8 - 15) \times 1$$

$$P = 4.8W$$

Ripple frequency:

$$f = 2 \times 50 = 100Hz$$

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IC Used: LM7815

Input capacitor = $0.33\mu\text{F}$

Output capacitor = $0.1\mu\text{F}$

Final regulated output:

$$V_{out} = 15V \text{ DC}$$

Ripple voltage formula:

$$V_r = \frac{I}{fC}$$

Rearranging:

$$C = \frac{I}{fV_r}$$

Assume allowable ripple voltage $V_r = 2V$

$$C = \frac{1}{100 \times 2}$$

$$C = 0.005F$$

$$C = 5000\mu F$$

Standard value selected:

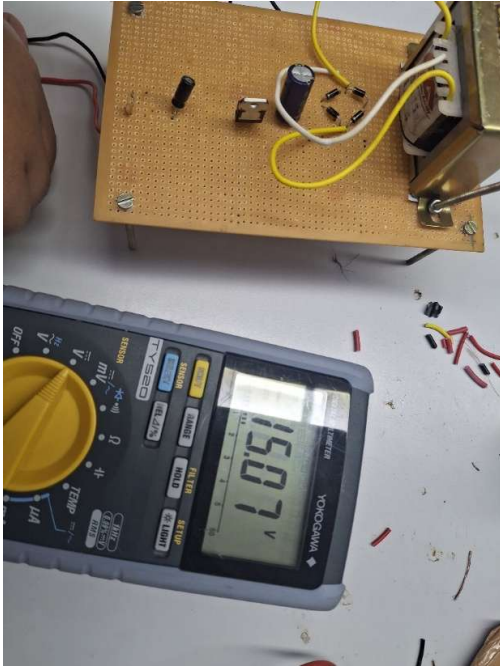
$$C = 4700\mu F \text{ or } 6800\mu F$$

Capacitor voltage rating $\geq 35V$

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Output results:



Conclusion and Results:

- Transformer output: 15 V AC
- Regulated output voltage: +15 V DC
- Ripple voltage was significantly reduced after using the filtering.
- The output voltage remained constant with load variation.

A regulated power supply converting 240 V AC to +15 V DC was successfully designed and simulated. The LM7815 voltage regulator provided a constant and ripple-free output voltage, making the circuit suitable for practical electronic applications.